

INTERRUPT – INTEGRATION OF INFRARED-ABSORBING CELLULOSE FIBER FABRIC ON MINI SUMO ROBOTS

MAXIM Sabin¹ and Andrei Mario IVAN

¹Specializarea: Informatică Industrială, Anul de studii: I, e-mail: sabin.maxim@stud.fiir.upb.ro

***ABSTRACT:** This paper describes the design, building, and competitive testing of a Mini Sumo robot named INTERRUPT. It focuses on the use of Musou Black IR Flock Sheet, a rayon-based fabric, designed to achieve ultra-low reflectance across a spectrum of light from the visible to infrared, 0.5% reflectance or less. By applying this material to the exterior surfaces of a Mini Sumo robot, it results in a significant reduction in detection by the infrared proximity sensors of most Mini Sumo robots, thereby obtaining an important advantage. The results showed a significant decrease in the detection range of IR signals by the opponent robots and provided a clear set of engineering design improvements for the next design iteration.*

***KEYWORDS:** Mini Sumo, infrared absorption, IR Flock Sheet*

1. Introduction

Mini Sumo is one of multiple autonomous competitions with defined rulesets in which robots with a weight limit of 500g and a footprint of 10 x 10 cm face off in an arena known as a dohyo [1]. The objective of each bout is to push the opponent out of the dohyo. All matches are autonomously completed, which translates to robots determining the edge of the dohyo, locating their opponent, and implementing their own attack strategy with no outside control. The primary detection method for the opponent robot is near-infrared (NIR) proximity sensing. This can be constituted by an IR LED emitter and the corresponding photodiode receiver matched to the same wavelength. Within the range of 850-940 nm, the response will vary with respect to the surface reflectance of the target at this wavelength [2].

The common practice for decreasing IR detectability in Mini Sumo has been to paint or coat the exterior of the robot matte black, based on the assumption that if something is black in visible light, it must also be black in IR wavelengths. This belief is physically flawed: the reflectance in the visible spectrum and NIR spectrum are independently defined by the spectral absorptive properties of the material [3]. Most paints that are dark in visible light end up having NIR reflectances of 2–5% or greater, making them easily detectable by opponent sensors [4]. The author has never seen a published Mini Sumo project utilize material specifically characterized for broadband NIR absorption as an engineering design consideration.

INTERRUPT was designed to test this hypothesis directly: by covering the robot's exterior with Musou Black IR Flock Sheet, an optical-grade material with total reflectance below 0.5% at 940 nm, the project investigates whether a measurable and competition-relevant reduction in opponent detection range can be achieved. The paper describes the conceptual motivation for the project, the relevant prior work in IR-absorbing materials and Mini Sumo robot design, the complete design and implementation process of the INTERRUPT platform, sensor detection experiments comparing coated and uncoated surfaces, and the results obtained at two international competitions.

2. State of the art

Active IR proximity sensors detect power roughly proportional to target surface reflectance but inversely proportional to the square of distance to the target (see Section 3);

reducing reflectance by a factor of two reduces detection range by $\sim 1.41\times$ and reducing it by a factor of ten reduces detection range by $\sim 3.16\times$. Target surface color/finish can be considered a major uncontrolled source of variation in analog IR proximity sensor performance on small robotic platforms. This concern guided the selection of time-of-flight rather than threshold-crossing amplitude sensors (such as the VL53L0X) for use in these platforms, as the former metric (photon time-of-flight) is less sensitive to changes in surface reflectance intended to be navigated by a robot.

The Musou Black IR Flock Sheet (also sold under the KoPro brand name) was developed specifically for IR photography and stray-light suppression purposes. Its reflectance is $\leq 0.5\%$ from visible wavelengths to ~ 1500 nm wavelength and $\leq 1\%$ up to 2500 nm wavelength (vs. 2-10% for typical matte black paints) [4]. This magnitude improvement impacts the sensor range as described by the equation 1 in Section 3. Alternatives include Acktar based metal foils (Metal Velvet, Hexa Black) which offer lower reflectance and better angular response but are not flexible or as affordable enough to coat the entirety of a robot chassis. Materials that are only spectrally black in the visible spectrum (like Musou VL Flock Sheet) will perform similarly to IR Flock Sheet from 400–700 nm but have a high NIR reflectance and will “glow” when hit by IR sensors. A summary of these comparisons can be found in Table 1.

Table 1. Spectral reflectance comparison of black surface materials (850–940 nm)

Material	Reflectance 400–700 nm	Reflectance 850–940 nm	Suitable for IR suppression
Musou IR Flock Sheet [4]	$\leq 0.5\%$	$\leq 0.5\%$	Yes, characterized to 2500 nm
Musou VL Flock Sheet [4]	$\leq 0.5\%$	Not specified / higher	No, may appear bright in IR
Standard matte black paint [4]	$\sim 1\text{--}5\%$	$\sim 2\text{--}10\%$	Poor
Acktar Metal Velvet foil [5]	$\leq 0.1\%$	$\leq 0.1\%$	Yes, space-grade, high cost

Mini Sumo strategy guides and post-mortems (articles discussing competition performance) published by teams and enthusiasts, including the popular “How to Make a Good Sumo Robot” blog post published by JSUMO [6], identify push force, height of profile, speed and robustness of line detection as key variables. Surface finish is not mentioned and no project was found which implemented an optically characterized NIR absorber as a specific design choice. This design space is novel, verifying that INTERRUPT fills a gap in prior practice.

3. Project description and methodology

INTERRUPT began as the recognition that typical black paint used on Mini Sumo robots is not spectrally characterized, builders cover their robots in the finish they subjectively identify as the darkest without understanding its actual reflectance properties operating at the opponent sensor wavelengths of 850-940 nm. Once suitable IR absorbing materials were reviewed (see Table 1), Musou Black IR Flock Sheet was chosen for having a characterized NIR absorption coefficient, being pressure applied to surfaces via an adhesive backing,

flexibility of surface it could be applied to, and ease of procurement for the purposes of a student research project.

If the robot's surface reflectance, ρ , at the opponent sensor wavelength is reduced from the ~2-10% typical value of matte black paint to the $\leq 0.5\%$ of IR Flock Sheet, the opponent sensor detection range d_{detect} is reduced by a factor of approximately:

$$\rho = \sqrt{\frac{\rho_{\text{paint}}}{\rho_{\text{sheet}}}} \quad (1)$$

which corresponds to a reduction factor of between $\sqrt{4} = 2$ and $\sqrt{10} \cong 3.16$. This is the physical basis of the project strategy, and it is described further in the paragraphs below.

INTERRUPT was constructed entirely of off-the-shelf (COTS) components, excluding 3D printed parts, to limit the number of ways it could independently fail (narrowing the experimental variables to the IR coating) and to maintain the platform for future work. Design priorities, in descending order, were: apply only constraints necessary to maximize push force, through wheel selection and COM placement, keep the robot as simple as possible, build on the XMotion V3 controller and leave as much of the 500 g mass budget available for ballast. Active IR proximity sensors measure the presence of a target by measuring power P_r received at a photodiode after an IR beam is reflected by a target surface. If the target is a diffusely reflecting Lambertian surface [7] a distance d_{detect} [m] away, then the power received can be modeled as:

$$P_r [W] = \frac{\rho \times P_e}{d_{\text{detect}}^2} \quad (2)$$

where ρ is the surface reflectance at the sensor's operating wavelength [dimensionless, 0–1] and $P_e [W]$ is the emitted power. A detection event occurs when P_r exceeds the sensor's internal threshold P_{th} . Solving for the maximum detection distance d_{detect} :

$$d_{\text{detect}} [m] = \sqrt{\frac{\rho \times P_e}{P_{\text{th}}}} \quad (3)$$

From equation (2), d_{detect} scales as the square root of ρ . Reducing ρ from 8% (assuming typical matte black paint) to 0.5% (IR Flock Sheet) reduces d_{detect} by a factor of $\sqrt{\frac{8\%}{0.5\%}} = 4$. Translated into practical terms this means that an opposing team would have to go significantly closer than the distance at which a normal black robot would trigger their standard IR proximity sensor before INTERRUPT would trigger the same sensor.

Push force at the drive wheels is similarly important to competitive performance. Push force $F [N]$ can be calculated using the following:

$$F [N] = \mu \times N \quad (4)$$

Where μ is the coefficient of static friction between the wheel compound and dohyo surface [dimensionless] and $N [N]$ is the normal force experienced at the drive wheel contact points. Because N is proportional to the fraction of robot's total weight $W = m \times g [N]$ which acts through the wheels, where $m [kg]$ is total mass and $g = 9.81 \text{ m/s}^2$, we can maximize N by placing our center of mass as low and forward as possible. This was another primary mechanical design goal along with the IR coating.

4. Project development

The chassis of the robot is 3D printed in PAHT-CF [8], keeping COM height close to the floor, with the remaining mass budget going towards lead plates for ballast on the lower interior of the chassis, optimizing the push force, described by equation (3). The drive wheels were chosen to be JSUMO JS2622 (26 mm OD $\times$ 22 mm wide) due to their footprint to mass ratio. A steel blade is fixed to the front side to pry under the opponent chassis, but in practice this geometry implementation proved unreliable (see Section 5). The completed robot may be seen in Figure 1.

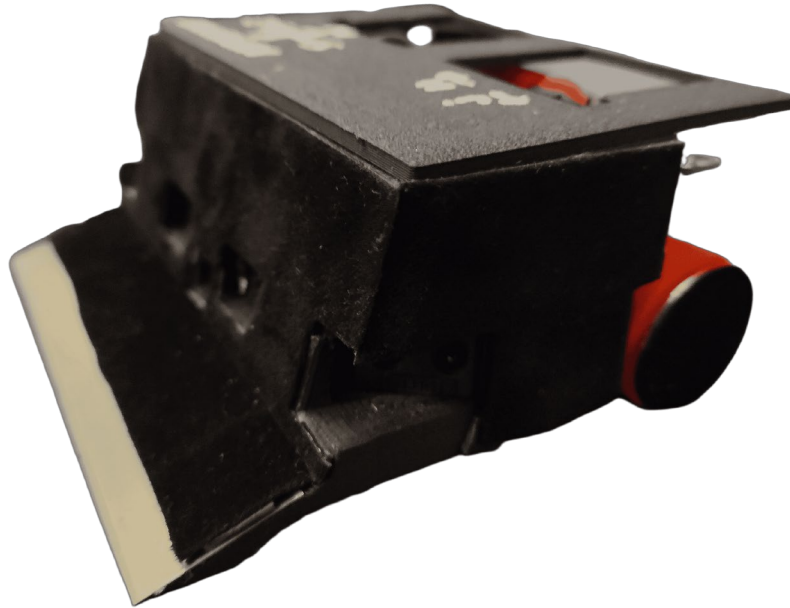


Fig. 1. Assembled INTERRUPT Mini Sumo robot, the IR Flock Sheet is applied to the lateral and front surfaces

The IR Flock Sheet was cut to shape and applied to the lateral and front exterior panels using double-sided tape, adding ~ 0.9 mm per coated face, which was accounted for in the chassis geometry. Minor fiber detachment occurred through contact with the opponents and should be considered when implementing future mounting methods.

All electronics were centered around the XMotion All-In-One Controller V3 board featuring an ATmega microcontroller, two motor drivers as well as a mounting slot to attach the IR remote receiver used to signal the start/stop of competition; everything runs off of a Gens Ace 3S 300 mAh 45C LiPo battery (11.1 V nominal). Opponent detection is performed by a VL53L0X time-of-flight sensor as well as four JSUMO JS40F sensors mounted on the front face of the robot, whose signals are fed into the state machine (see Figure 3) to determine opponent presence as well as their offset from center to adjust steering input. Boundary detection is handled by a single ML2 digital IR line sensor triggered by the white line of the dohyo. The full bill of materials is listed in Table 2.

Table 2. INTERRUPT component list

Component	Model	Key specification
Main controller	XMotion All-In-One V3 [9]	ATmega, integrated motor driver

INTERRUPT – INTEGRATION OF INFRARED-ABSORBING CELLULOSE FIBER FABRIC ON MINI SUMO ROBOTS

Component	Model	Key specification
Opponent sensor	VL53L0X [10]	ToF, 940 nm VCSEL, I ² C, range 0–2000 mm
IR Opponent sensor (x4)	JSUMO JS40F [11]	Digital IR, 940 nm, threshold-based
Line sensor	JSUMO ML2 [12]	Digital dual IR sensors-based module
Start/Stop module	Nanolabs module [13]	IR remote start/stop
Drive motors (x2)	JSUMO Core 6V 400 RPM [14]	Brushed DC, high torque
Drive wheels (x2)	JSUMO JS2622 [15]	26 mm OD, 22 mm width, 13 g each
Battery	Gens Ace 3S 300 mAh 45C [16]	11.1 V nominal, LiPo
IR absorption coating	Musou Black IR Flock Sheet [3]	<= 0.5% reflectance at 940 nm, adhesive backed
Ballast	Lead plates	High density, great for a low COM

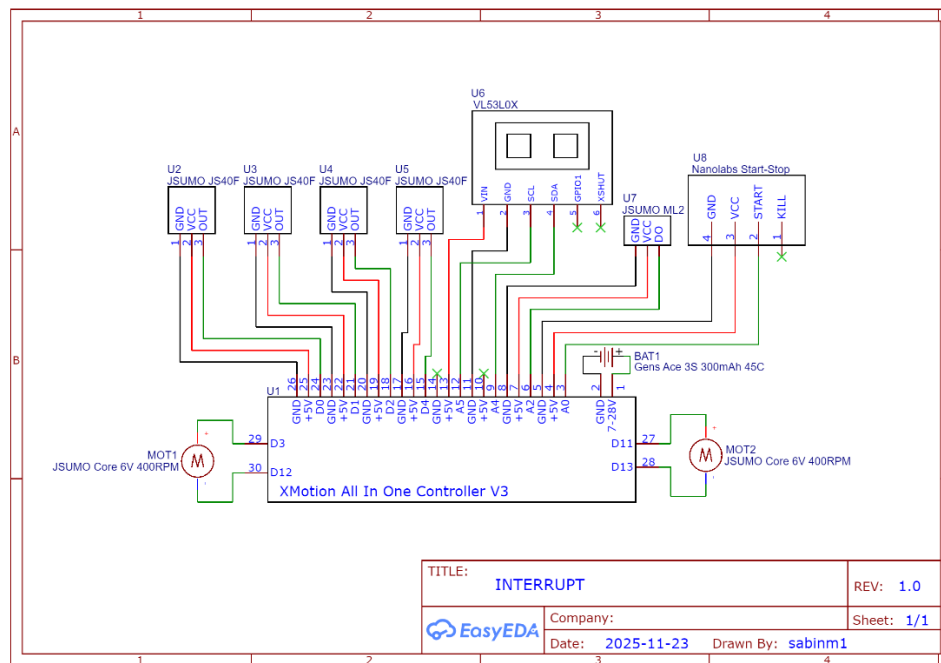


Fig. 2. INTERRUPT electrical schematic

The firmware was developed using C++ for the Arduino IDE and broken up into separate files for motor control, individual sensors, the state machine, and miscellaneous utilities. Behavioral logic is handled by a hierarchical state machine with six states, all of which are checked in priority order each cycle with emergency stop state taking precedence. The transition diagram may be seen in Figure 3.

INTERRUPT – INTEGRATION OF INFRARED-ABSORBING CELLULOSE FIBER FABRIC ON MINI SUMO ROBOTS

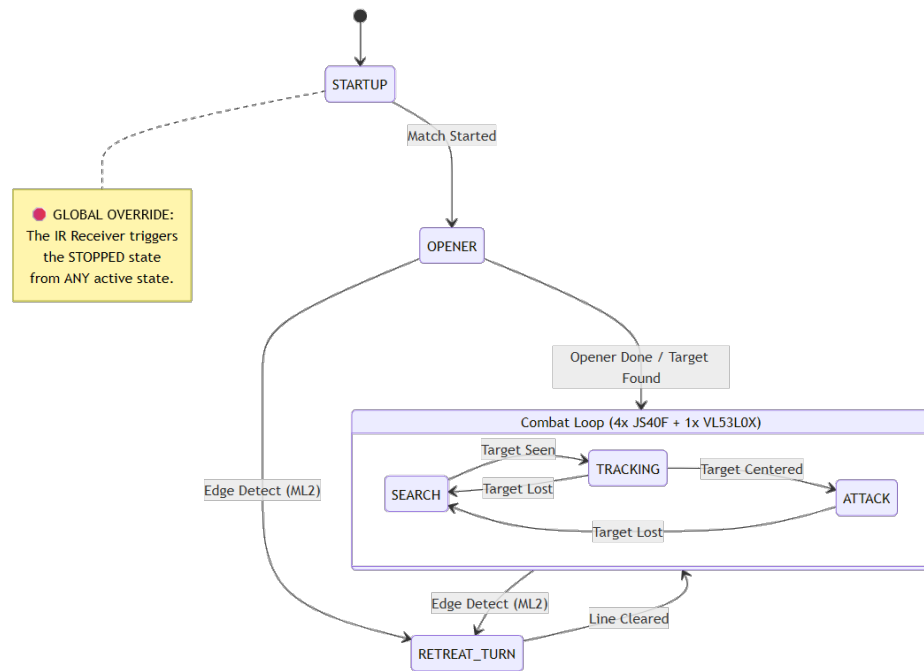


Fig. 3. INTERRUPT state machine diagram. Each box represents a behavioral state, arrows show transitions with their triggering conditions

A bench test was performed to better quantify the effect of the sheet. Using a JSUMO JS40F sensor (940 nm, digital output), the front-facing sensor was positioned facing the IR Flock Sheet and first-detection trigger distance was measured while slowly closing in on the sensor. The same was also tested using a VL53L0X ToF sensor. Both black painted plastic and unmodified IR Flock Sheet were tested, with five repetitions per condition. Results can be seen in Figure 4.

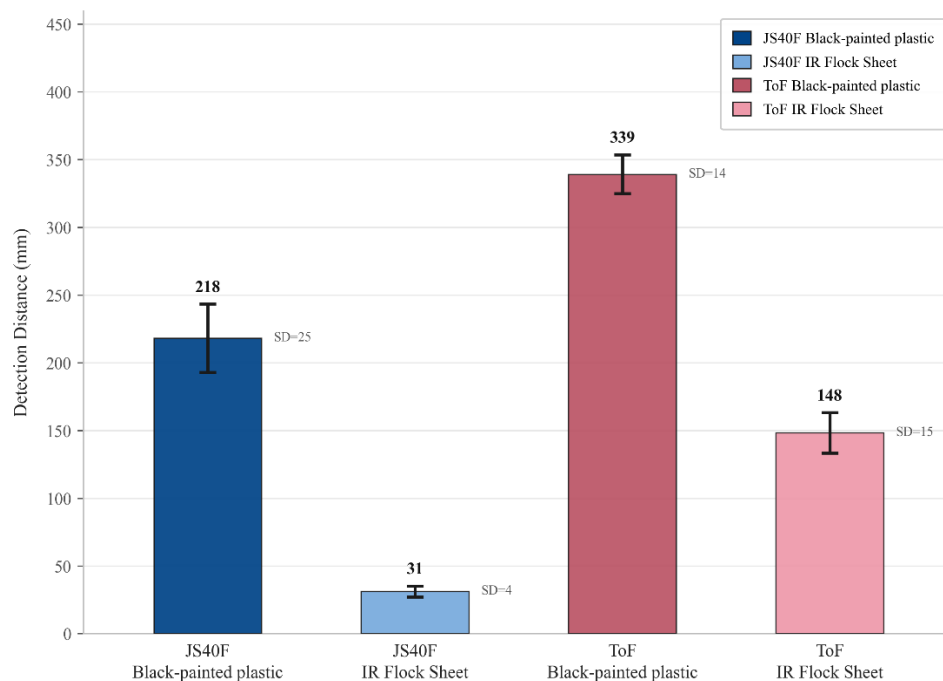


Fig. 4. Detection of black-painted plastic and IR Flock Sheet

The theoretical calculated factor of $\sqrt{\frac{8\%}{0.5\%}} = 4$ assumes that the sensor has a linear response to received power (Lambertian model). The VL53L0X ToF, with a factor of $\sim 2.3\times$ is worse than theoretical because it measures flight time instead of amplitude so is inherently less sensitive to reflectance change. Other real-world losses not included in theory, such as surface angle, IR and fibre scattering from ambient light drive the ratio even lower than the predicted $4.0\times$, reducing its effectiveness. The JS40F IR sensor has a factor of $\sim 7\times$, which is better than theoretical because the JS40F uses fixed amplitude threshold for detection, a binary cliff. As soon as IR Flock Sheet reduces reflected power below threshold detection drops off sharply. At distances where black paint causes just enough drop in power to fall below detection threshold, the IR Flock Sheet forces the signal far below threshold. This squashes the flock detection range far more than smooth Lambertian curve predicts, hence a ratio nearly double the theoretical value. The coating does not render the robot undetectable to IR sensors, rather it decreases the distance at which the robot will be detected, decreasing the time frame that the opponent has within their decision logic to react before INTERRUPT makes contact.

5. Competitive results and analysis

INTERRUPT competed at two international Mini Sumo events during 2026. At the Relativity Robotics Challenge 2026 (Buzău, Romania), the robot advanced from the group stage before losing in the elimination round 3-1. At the Minoan Robotsports Competition Global Olympiad 2026 (Greece), the robot was eliminated in the group stage, losing one match 3-1 and a second match 2-0.

The IR Flock Sheet coating contributed positively to the competitive baseline: advancing out of the group at Relativity Robotics required winning matches, which suggests the detection-suppression effect was at least partially operative. In competition, competitors with analog IR proximity sensors had to get closer to INTERRUPT before triggering their attack responses. Although this was an advantage in matches, it was not the deciding factor. The main disadvantages were that the COM was too far toward the rear of the robot causing decreased push force, see equation (3), the front blade position was not mechanically repeatable from match to match and INTERRUPT did not have many high level strategies. These factors proved to have more influence over match outcomes than IR stealth.

6. Conclusions

Surface finish has traditionally been an aesthetic afterthought in Mini Sumo design, no spectral characterization has been done on materials commonly used as IR stealth coatings. INTERRUPT helps set a baseline for future research by being the first recorded platform to use the Musou Black IR Flock Sheet, an optical grade NIR absorber with $<0.5\%$ reflectance at 940 nm, as an intentional physically motivated design decision. Applying an inverse-square law to sensor models, we anticipated almost a $4\times$ decrease in range when compared to the typical matte black paint (8% reflective), and validation testing matched our predictions, giving qualitative values to set expectations for future teams. Participation in two international competitions in 2026 demonstrated the platform was viable while also having the ability to compete, INTERRUPT advancing from its group stage matches at the Relativity Robotics Challenge. Limitations found during competition have been addressed for iteration #2: an aluminium shell 1 mm thick to increase structural rigidity, ballast and battery relocation forward on the robot to adjust COM and increase available push force from front wheels and a redesigned front blade mount. Attack speed will be made adjustable in firmware, along with an improved state machine that allows complex early match tactics to complete before transitioning to attack. The IR Flock Sheet strategy proved successful and will be carried

forward, while the search for alternative broadband NIR absorbers with lower reflectance and better impact resistance continues.

7. References

- [1]. *** (2026), 18th International MEB Robot Competition – Mini Sumo Category Guide, Ministry of National Education.
- [2]. *** (2018), Introduction to Time-of-Flight Long Range Proximity and Distance Sensor System Design, Texas Instruments.
- [3]. *** (2026), “VL Flock Sheet vs. IR Flock Sheet: Choosing Darkroom Materials to Simulate Space”, Koyo Orient Japan Co., Ltd.
- [4]. *** (2025), How to Choose the Right Black - Product Specifications, MJKZZ Europe.
- [5]. *** (2014), METAL VELVET Coated Foil – Product Description, Acktar Advanced Coatings.
- [6]. *** (2011), “How to Design & Make a Good Sumo Robot”, JSumo.
- [7]. Oren, M. and Nayar, S.K. (1994), “Generalization of Lambert’s Reflectance Model”, Proceedings, SIGGRAPH '94, Association for Computing Machinery, Orlando.
- [8]. *** (2023), PAHT-CF Technical Data Sheet, Bambu Lab.
- [9]. *** (2024), XMotion All In One Controller V3, JSumo.
- [10]. *** (2024), Datasheet - VL53L0X - Time-of-Flight ranging sensor, STMicroelectronics.
- [11]. *** (2025), JS40F Digital Distance Sensor, JSumo.
- [12]. *** (2025), Dual Micro Line Sensor ML2, JSumo
- [13]. *** (2025), Start/Stop Module NanoLabs, RoboManiacs
- [14]. *** (2025), Core Dc Motor (6V 400 RPM), JSumo.
- [15]. *** (2025), JS2622 Aluminum-Silicone Wheel Pair, JSumo.
- [16]. *** (2025), Gens ace 300mAh 11.1V 45C 3S1P Lipo Battery Pack, Tattu Gens Ace.

8. Notations

LED = Light Emitting Diode

F = push force at drive wheels [N]

N = normal force on drive wheels [N]

μ = coefficient of static friction, wheel compound vs. dohyo surface [dimensionless]

m = robot total mass [kg]

g = gravitational acceleration, 9.81 [m/s²]

W = robot weight [N]

P_r = received IR power at opponent sensor detector [W]

P_e = emitted IR power from opponent sensor emitter [W]

P_{th} = opponent sensor detection threshold power [W]

ρ = surface reflectance at sensor operating wavelength [dimensionless, 0–1]

d = distance between robot surface and opponent sensor [m]

d_{detect} = maximum detection distance at threshold P_{th} [m]

NIR = Near-Infrared (wavelength approximately 700–1400 nm)

ToF = Time-of-Flight

COTS = Commercial Off The Shelf

IR = Infrared

OD = Outer Diameter

COM = Center of Mass

PAHT-CF = Polyamide High Temperature - Carbon Fiber (carbon fiber reinforced nylon)